CI/CD Pipeline for a Containerized Java Calculator

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1 Introduction

Modern software systems are rarely developed or deployed in isolation. Instead, they rely on a coordinated ecosystem of tools that automate building, testing, packaging, and deployment. The discipline that orchestrates these processes is commonly referred to as DevOps. DevOps integrates development and operational practices in order to facilitate continuous integration and continuous deployment (CI/CD), thereby improving both the reliability and velocity of software delivery.

The objective of this project is to construct a minimal yet functional CI/CD pipeline for a command-line scientific calculator implemented in Java. The pipeline automates compilation, testing, containerization, and deployment through a sequence of interconnected tools.

The primary technologies employed in this project are:

- **GitHub** – A distributed version control platform used for source code hosting and collaboration. It serves as the canonical repository for the calculator application's source code.
- **Jenkins** – An open-source automation server responsible for executing the CI/CD pipeline. Jenkins retrieves the source code from GitHub, runs automated tests, builds the project, and orchestrates subsequent steps.
- **Docker** – A containerization platform used to package the calculator application and its runtime environment into a portable container image.
- **Docker Hub** – A container registry used to store and distribute Docker images. The built image is pushed to Docker Hub and later retrieved during deployment.
- **Ansible** – A configuration management and automation tool used to provision the deployment environment and execute container-based deployment tasks.

The combination of these tools establishes a streamlined pipeline that automatically transforms source code into a deployable application.

The project repositories are available at:

- GitHub Repository:
  `https://github.com/VenerableHarsha/Miniproject-calculator.git`
- Docker Hub Image Repository:
  `https://hub.docker.com/repository/docker/venerableharsha/calculator-app/general`

2 CI/CD Pipeline with Jenkins

Jenkins functions as the central orchestration engine of the pipeline. Upon receiving updates from the GitHub repository, Jenkins initiates a series of automated stages defined in the `Jenkinsfile`. These stages collectively perform compilation, testing, container image construction, and deployment.

The pipeline consists of the following stages:

- **Source Retrieval** – Jenkins clones the calculator repository from GitHub using the Git SCM plugin.
- **Build and Compilation** – The project is compiled using Apache Maven. Maven resolves dependencies and produces a runnable JAR file.
- **Testing** – Unit tests implemented with JUnit are executed. This stage ensures that the calculator's mathematical operations behave correctly prior to containerization.
- **Container Image Construction** – Docker is used to construct a container image containing the compiled calculator application.
- **Image Publication** – The resulting Docker image is pushed to Docker Hub, making it accessible for deployment.
- **Deployment** – Ansible is invoked to provision the deployment environment and launch the calculator container.
- **Post-Action** - Jenkins triggers an email notification to notify upon the build success or failure

3 Containerization with Docker

Docker is employed to encapsulate the application and its runtime environment within a container image. This approach ensures that the application executes consistently across different environments.

The Dockerfile defines the steps required to build the container image. The base image provides a lightweight Java runtime environment, upon which the compiled calculator application is placed.

Once constructed, the image is tagged and pushed to Docker Hub. This enables the deployment stage to retrieve the image from a centralized registry rather than relying on locally built artifacts.

3.1 Jenkins Pipeline Configuration

After installing Jenkins, a new pipeline job was created to automate the build and deployment process for the calculator application.

From the Jenkins dashboard, the option **New Item** was selected and a new project was created using the **Pipeline** project type. This option allows Jenkins to execute build instructions defined in a `Jenkinsfile` stored in the project's source repository.

The pipeline job was configured to retrieve the project source code directly from the GitHub repository. The repository URL was provided in the pipeline configuration so that Jenkins could clone the project during execution.

Additionally, the pipeline definition was configured to use the **Pipeline Script from SCM** option. This allows Jenkins to automatically load the pipeline instructions from the `Jenkinsfile` present in the repository.

The following screenshots illustrate the configuration steps used to create the Jenkins pipeline job.

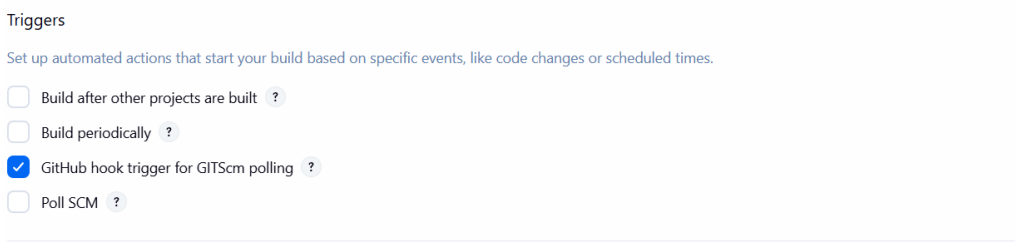


Figure 1: Creating a new Jenkins pipeline project

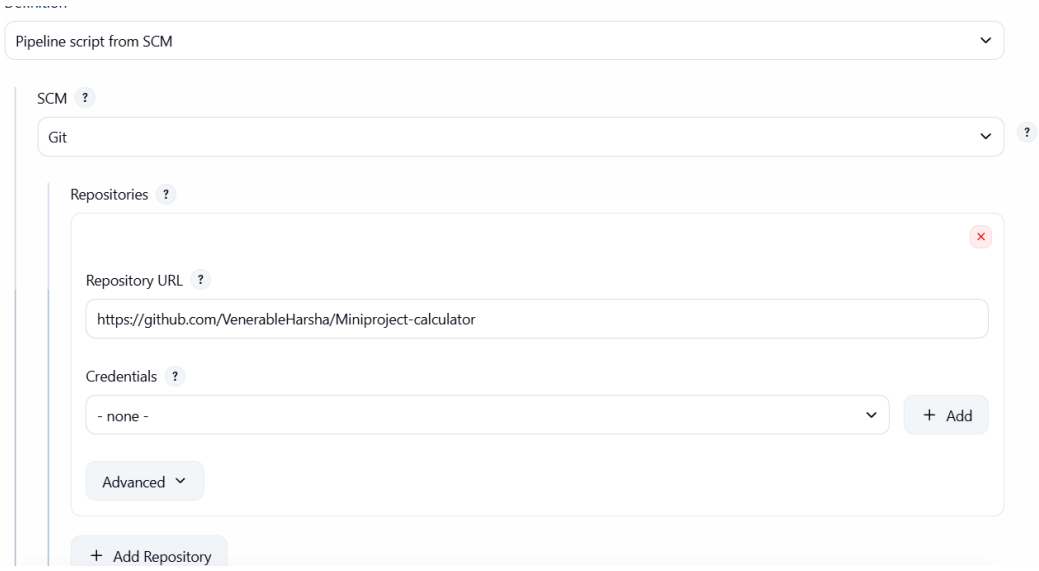


Figure 2: Jenkins pipeline configuration using the GitHub repository

4 Deployment Automation with Ansible

Ansible is utilized to automate the provisioning of the deployment environment and the execution of container operations.

The `inventory.ini` file specifies the target host for deployment. In this project, the host corresponds to the local deployment environment managed within the Docker ecosystem.

The `deploy.yml` playbook contains the automation logic. Its responsibilities include:

- Ensuring that the deployment container exists and remains active.
- Pulling the latest calculator image from Docker Hub.
- Launching the calculator container within the deployment environment.

Through this mechanism, the deployment process becomes reproducible and consistent, eliminating the need for manual configuration.

5 System Architecture

The complete CI/CD workflow integrates several tools into a coherent pipeline. Source code modifications trigger automated build and deployment processes that ultimately produce a runnable containerized application.

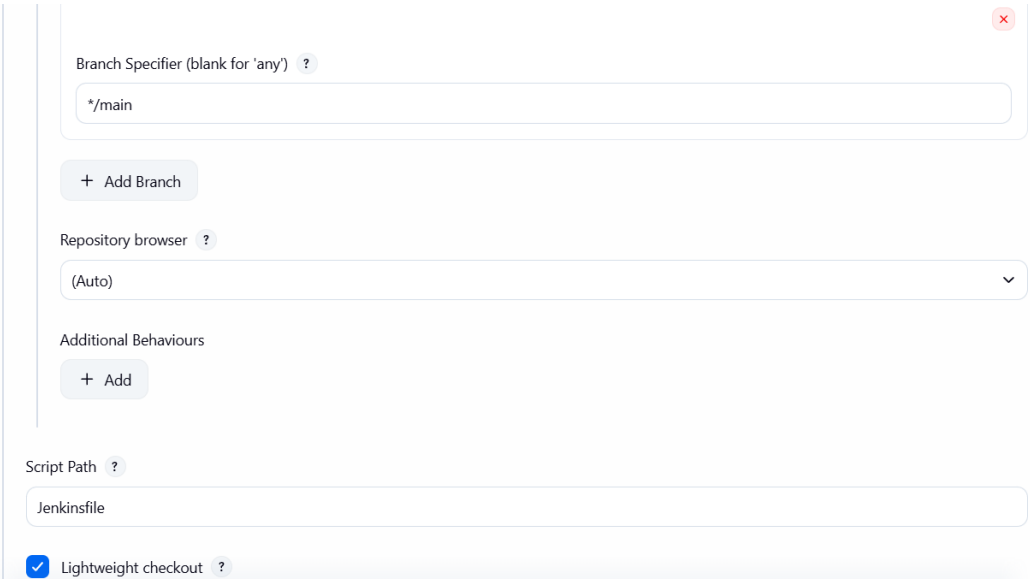


Figure 3: Pipeline configured to use Jenkinsfile from the source repository

The high-level architecture of the system is illustrated using the PlantUML diagram below.

Figure 4: Architecture diagram

5.1 Development and Execution Environment

All development and pipeline execution in this project was performed on a single local workstation using Windows and Windows Subsystem for Linux (WSL).

The development workflow was divided across multiple environments:

- **Development Environment (Windows)**
  The calculator application was developed using IntelliJ IDEA running on Windows. Source code editing, project structure management, and local testing were performed within this environment.

- **Build and CI Environment (WSL Ubuntu)**
  Jenkins, Docker, Maven, and Ansible were installed inside a WSL Ubuntu environment. This Linux environment served as the CI/CD execution platform responsible for compiling the application, running tests, building container images, and executing deployment automation.

- **Deployment Environment (Fresh Ubuntu Container)**
  For deployment simulation, a fresh Ubuntu container was created to act as the target host. This container represented a clean runtime environment into which the application container would be deployed.

To reduce resource overhead, a separate Docker daemon was not launched inside the deployment container. Instead, the container was configured to use the Docker service already running in the WSL host environment. This approach allowed the deployment process to interact with Docker without running nested container services.

5.2 Exposing the Jenkins Server using ngrok

Since Jenkins was running locally inside the WSL Ubuntu environment, it was not directly accessible from external services such as GitHub. However, the CI/CD pipeline relies on GitHub webhooks to trigger builds whenever changes are pushed to the repository.

To enable GitHub to communicate with the locally running Jenkins server, **ngrok** was used to expose the Jenkins HTTP endpoint to the public internet.

Ngrok creates a secure tunnel from a public URL to a local service. In this project, the Jenkins server running on port 8080 was exposed using the following command:

```
ngrok http 8080
```

This command generated a temporary public URL that forwarded incoming HTTP requests to the Jenkins server running on the local machine.

The generated ngrok URL was then used to configure the GitHub webhook so that any push event in the repository would trigger the Jenkins pipeline.

Through this mechanism, GitHub was able to notify Jenkins of repository updates even though Jenkins was running within a local development environment.

A screenshot of the active ngrok tunnel is shown below.

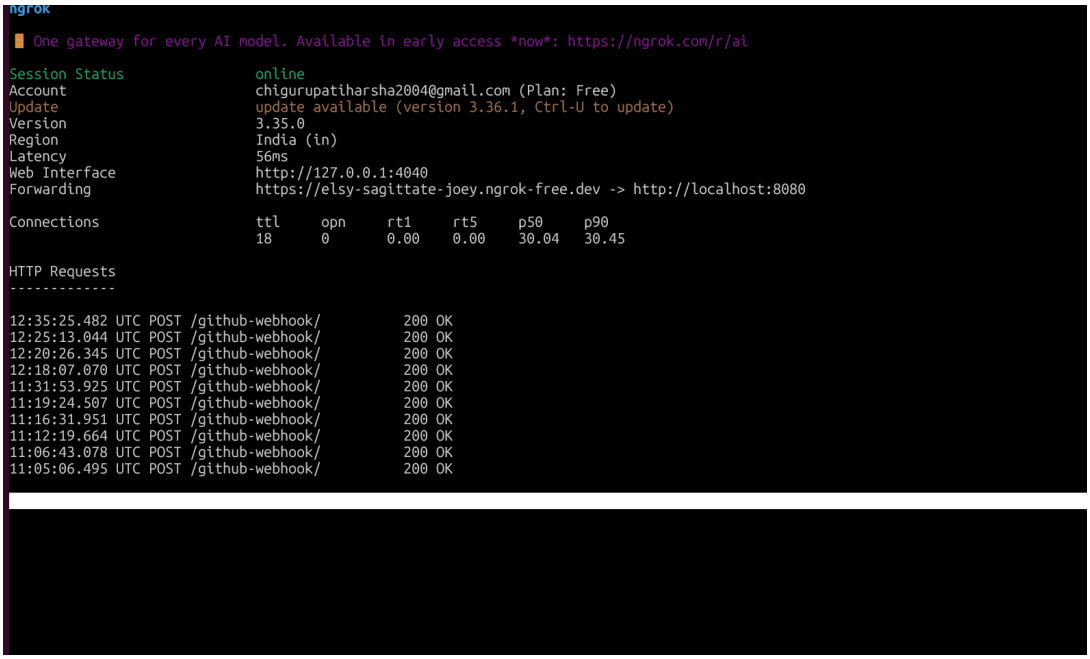


Figure 5: Running Ngrok that catches webhooks

5.3 Jenkins Pipeline Execution Stages

The Jenkins pipeline is defined through a declarative configuration file known as the `Jenkinsfile`. This file specifies a sequence of stages that are executed whenever the pipeline is triggered.

Each stage performs a specific task in transforming the source code into a deployed containerized application. The following sections describe the function of each stage in the pipeline.

5.3.1 Build Stage

The pipeline begins with the build stage. In this step Jenkins navigates to the project directory containing the calculator source code.

The Apache Maven build system is then invoked to compile the Java source files. During this process Maven performs several tasks automatically:

- Downloads any required dependencies from the Maven Central repository if they are not already present locally.
- Compiles all Java source files into bytecode.
- Verifies that the project structure conforms to the Maven build model.

At the end of this stage the project has been successfully compiled but has not yet been packaged into a runnable artifact.

5.3.2 Test Execution Stage

After compilation completes successfully, the pipeline proceeds to the testing stage. In this stage Maven executes the project's automated unit tests.

The tests validate the correctness of the calculator's mathematical operations such as addition, subtraction, multiplication, and division. Running these tests before packaging ensures that only verified code proceeds further in the pipeline.

If any test fails, the Jenkins pipeline immediately terminates and the build is marked as failed. This prevents defective code from being packaged or deployed.

If all tests pass successfully, Jenkins advances to the packaging stage.

5.3.3 Application Packaging Stage

Once testing has completed successfully, the pipeline packages the compiled application.

Maven produces a runnable Java archive (JAR file) containing the compiled calculator application and the metadata required to execute it. This artifact serves as the executable component that will later be placed inside the Docker container image.

The packaging step ensures that the application exists as a single portable binary which can be easily transferred between environments.

5.3.4 Docker Image Construction

After the application has been packaged, the pipeline constructs a Docker container image.

During this step Docker reads the instructions contained in the project's Dockerfile. The Dockerfile defines the base runtime environment, copies the packaged application artifact into the container, and specifies the command required to launch the calculator program.

The result of this process is a Docker image containing the calculator application together with the Java runtime environment necessary to execute it.

This image is tagged using a version identifier so that it can later be published to a container registry.

5.3.5 Docker Image Publication

Once the container image has been built locally, the pipeline publishes the image to Docker Hub.

To perform this operation Jenkins authenticates with Docker Hub using stored credentials configured within the Jenkins credential manager. These credentials allow Jenkins to securely log into the registry without exposing authentication information in the pipeline configuration.

After authentication, the image is pushed to Docker Hub. This step makes the container image available to external systems and deployment environments.

Using a centralized registry ensures that deployment processes retrieve a consistent and verified version of the application.

5.3.6 Deployment using Ansible

The final stage of the pipeline performs automated deployment using Ansible.

In this stage Jenkins invokes an Ansible playbook which contains the instructions required to deploy the application container. The playbook interacts with the deployment environment specified in the `inventory.ini` configuration file.

During deployment the playbook performs the following actions:

- Connects to the target deployment host.
- Pulls the latest calculator container image from Docker Hub.
- Starts a new Docker container running the calculator application.

The deployment process therefore retrieves the container image produced by the pipeline and launches it in a fresh runtime environment. This simulates the behavior of a production deployment pipeline where container images are distributed and executed across remote hosts.

Finally we get an email notification, for this project I have used dummy email as sender and a non-existent email as reciever.

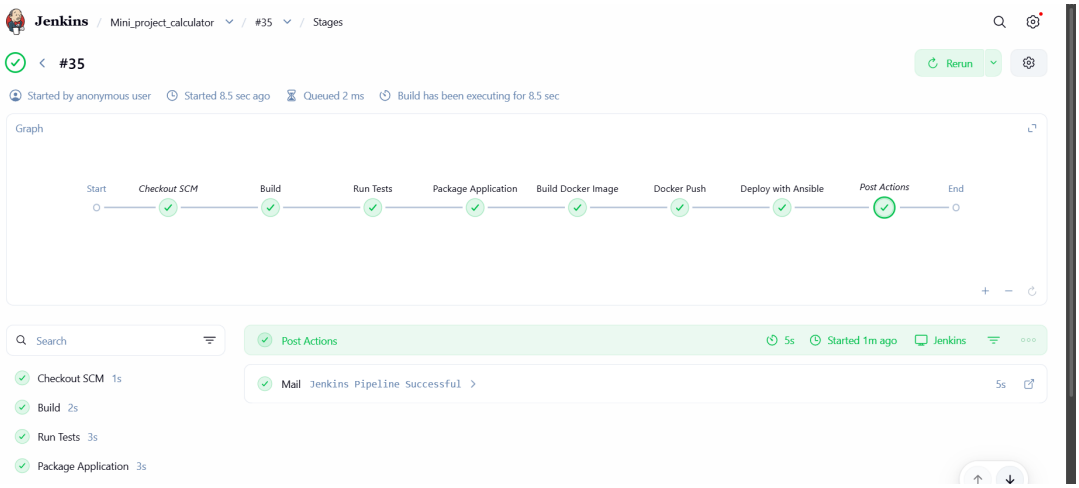


Figure 6: A successful pipeline build

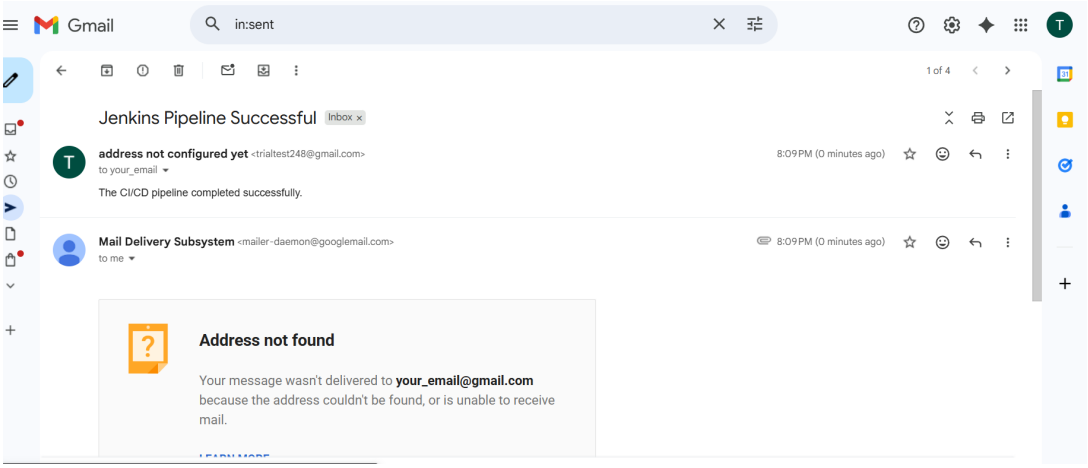


Figure 7: Example email notification

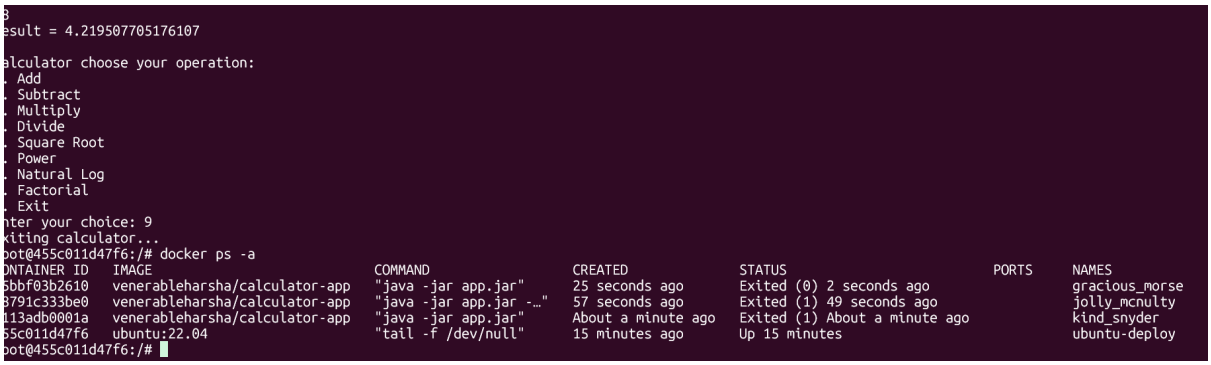
6 Conclusion

This project demonstrates the construction of a lightweight yet fully functional CI/CD pipeline integrating several widely adopted DevOps technologies. By combining version control, automated builds, containerization, and configuration management, the pipeline ensures that application updates can be propagated from source code to deployment with minimal manual intervention.

The architecture illustrates how modern software delivery pipelines leverage automation to improve reliability, reproducibility, and operational efficiency.

7 References

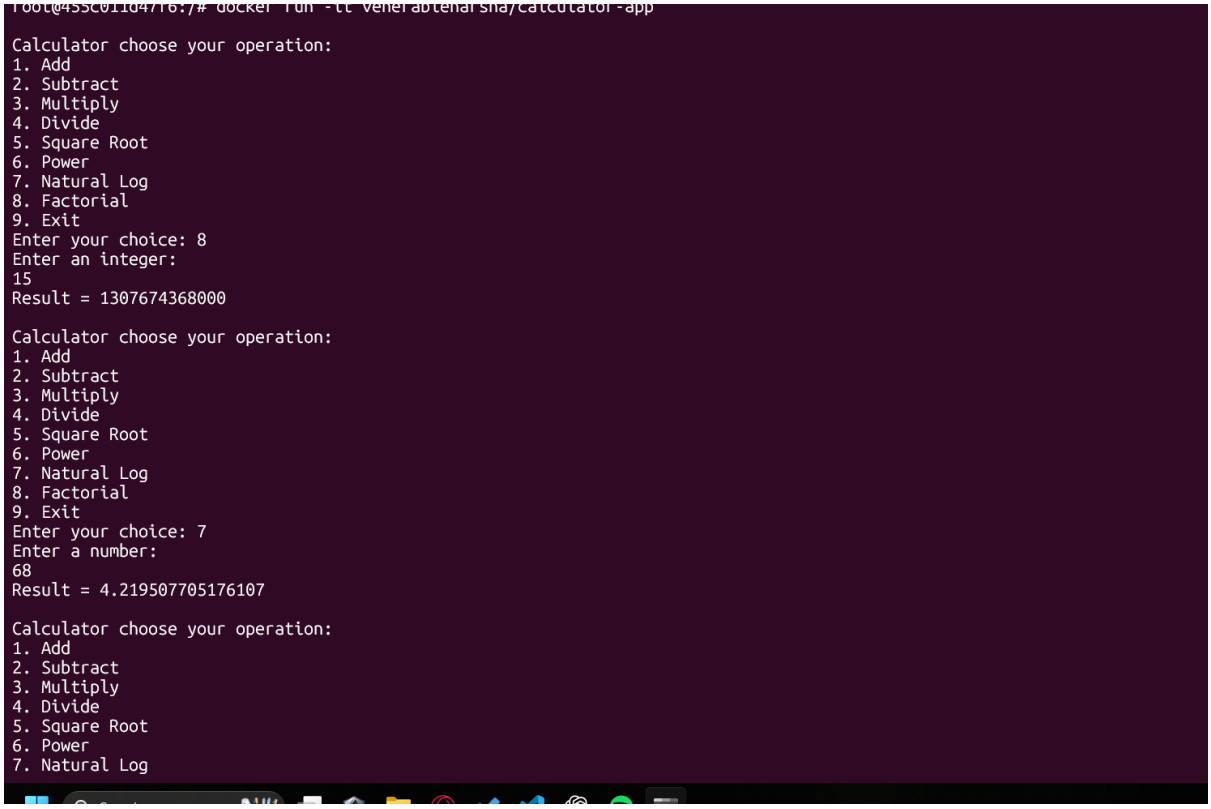
- Docker Documentation – https://docs.docker.com
- Jenkins Documentation – https://www.jenkins.io/doc/
- Ansible Documentation – https://docs.ansible.com
- Maven Documentation – https://maven.apache.org



```
8
esult = 4.219507705176107

alculator choose your operation:
. Add
. Subtract
. Multiply
. Divide
. Square Root
. Power
. Natural Log
. Factorial
. Exit
nter your choice: 9
xiting calculator...
ot@455c011d47f6:/# docker ps -a
ONTAINER ID   IMAGE                          COMMAND                  CREATED              STATUS                          PORTS     NAMES
6bbf03b2610   venerableharsha/calculator-app   "java -jar app.jar"      25 seconds ago       Exited (0) 2 seconds ago                  gracious_morse
8791c333be0   venerableharsha/calculator-app   "java -jar app.jar -…"   57 seconds ago       Exited (1) 49 seconds ago                 jolly_mcnulty
113adb0001a   venerableharsha/calculator-app   "java -jar app.jar"      About a minute ago   Exited (1) About a minute ago             kind_snyder
55c011d47f6   ubuntu:22.04                   "tail -f /dev/null"      15 minutes ago       Up 15 minutes                             ubuntu-deploy
ot@455c011d47f6:/#
```

Figure 8: Docker images once deployed



```
root@455c011d47f6:/# docker run -it venerableharsha/calculator-app

Calculator choose your operation:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Square Root
6. Power
7. Natural Log
8. Factorial
9. Exit
Enter your choice: 8
Enter an integer:
15
Result = 1307674368000

Calculator choose your operation:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Square Root
6. Power
7. Natural Log
8. Factorial
9. Exit
Enter your choice: 7
Enter a number:
68
Result = 4.219507705176107

Calculator choose your operation:
1. Add
2. Subtract
3. Multiply
4. Divide
5. Square Root
6. Power
7. Natural Log
```

Figure 9: Runnning calculator